

University of Dhaka
CSE 2106: Microprocessor and Assembly Language Lab
Lab Evaluation - 1
Date: 10.10.2025

1. Write an assembly language code to perform circular bit rotation of a 32-bit integer by n positions to the left. Store the result in memory.

Sample Input:

Value at 0x20000000: 0x12345678

Rotate left by n = 8 bits (value stored at 0x20000004)

Sample Output:

Result at 0x20000008: 0x34567812

2. Write an assembly code to implement emergency braking logic using sensor inputs and conditional branching.

Inputs:

SPEED (0x20000000): 32-bit

DISTANCE (0x20000004): 32-bit

BRAKE_PRESSURE (0x20000008): 32-bit

Output:

MODE (0x2000000C): 0=Idle, 1=Normal Brake, 2=Warning, 3=Emergency Brake

Conditions:

SPEED > 80 and DISTANCE < 100 → MODE = 3

DISTANCE < 100 → MODE = 2

BRAKE_PRESSURE > 80 → MODE = 1

Else → MODE = 0